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### **1. What is CSS Box Model?**

The CSS Box Model is a concept that describes how every HTML element is represented as a rectangular box.

It consists of four parts:

Content – The actual text or image inside the element.

Padding – Space between the content and border.

Border – A line surrounding the content and padding.

Margin – Space outside the border that separates the element from other elements.

### **2. What is Media Query? Explain with a sample code.**

A media query is a CSS technique used to apply different styles to a webpage based on the screen size, device, or other conditions.

It is mainly used for creating responsive web designs.

Example:

```
@media (max-width: 600px) {  
  body {  
background-color: lightblue;  
  }  
}
```

### **3. What are the features of Bootstrap?**

The main features of Bootstrap are:

Responsive design – Websites adjust to different screen sizes.

Grid system – Provides a 12-column layout system.

Predefined CSS classes – Provides ready-to-use styling.

Ready-made components – Includes buttons, cards, forms, navbars, etc.

JavaScript components – Provides interactive components.

Cross-browser compatibility – Supports modern web browsers.

Easy to use – Reduces the amount of CSS code required.

Mobile-first approach – Designed to work well on mobile devices.

### **4. Write an example of basic grid structure in Bootstrap.**

Bootstrap uses a 12-column grid system. A basic grid structure contains a container, row, and columns.

Example:

```
<div class="container">
<div class="row">
<div class="col-4">Column 1</div>
<div class="col-4">Column 2</div>
<div class="col-4">Column 3</div>
</div>
</div>
```

## 5. What is CSS Selectors?

CSS selectors are used to select HTML elements to which CSS styles are applied.

Common types of CSS selectors are:

Element selector

Class selector

ID selector

Universal selector

Grouping selector

Attribute selector

Pseudo-class selector

Pseudo-element selector

Descendant selector

Child selector

## 6. What are CSS backgrounds? List the properties.

CSS background properties are used to control the background of an HTML element.

Important background properties are:

background-color

background-image

background-repeat

background-position

background-size

background-attachment

background-origin

background-clip

background

## 7. How can we overlap elements in CSS?

Elements can be overlapped in CSS by using positioning properties such as relative, absolute, or fixed.

The z-index property can be used to control which overlapping element appears in front of another.

A higher z-index value generally places an element above an element with a lower value.

## 8. What are the different positioning properties in CSS?

There are five main CSS positioning properties:

Static – Default positioning of an element.

Relative – Positions an element relative to its normal position.

Absolute – Positions an element relative to its nearest positioned ancestor.

Fixed – Positions an element relative to the browser window and keeps it fixed during scrolling.

Sticky – Combines features of relative and fixed positioning depending on the scroll position.

## 9. What is ECMAScript?

ECMAScript is a standard or specification that defines the rules and features of scripting languages.

JavaScript is the most widely used programming language that implements the ECMAScript standard.

ECMAScript defines features such as variables, data types, functions, objects, classes, operators, loops, and other language features.

## 10. What is the difference between var, let and const in JavaScript? Explain Temporal Dead Zone.

Temporal Dead Zone (TDZ)

The Temporal Dead Zone is the period between entering a block and the declaration of a let or const variable.

During this period, the variable exists in the scope but cannot be accessed before its declaration.

Accessing it results in a ReferenceError.

Important: var does not have a Temporal Dead Zone in the same way because it is initialized with undefined during hoisting.

| Feature            | var            | let                       | const                     |
|--------------------|----------------|---------------------------|---------------------------|
| Scope              | Function scope | Block scope               | Block scope               |
| Redeclaration      | Allowed        | Not allowed in same scope | Not allowed in same scope |
| Reassignment       | Allowed        | Allowed                   | Not allowed               |
| Hoisting           | Yes            | Yes                       | Yes                       |
| Temporal Dead Zone | No             | Yes                       | Yes                       |